

Exercise 1: Moonbase

Goals/Purpose:

Team members must share information with each other. Members must gather and consolidate critical information and act on it, under a limitation of time and communication. All members must know the plan and understand the group goals to succeed. Tests individuals and group problem-solving and communication skills.

Group size:

11-30 members (must have at least 11 participants)

Time:

1 - 1.5 hours

Material:

- Clue ships
- Notepad
- Participant handout
- Markers to identify “bubble areas”
- Stopwatch or watch to keep time with

Process:

- *Planning period -- approx. 45 minutes*
- Pass out the handouts (have enough copies so all can see, or make overhead projector transparencies and project the handout)
- Observe how the students make their plan for the execution phase. Note key players and key points of the proposed plan(s).
- At the end of the allotted time (or when students say they’re finished) collect the handouts
- *Execution phase -- approx. 15-35 minutes*
- Hand out the clues, one per cadet. There are eleven critical clues, each having a serial number ending in “II”. All eleven of these clues must be given out to solve the exercise. Have the participants go to their appropriate “bubble”. They may not show their clue to anyone since it represents “head” knowledge. Also, explain that the time begins when you say “begin”.
- When the participants are in their ‘bubble’ areas, give the start signal and start the timer
- Observe the cadets’ execution of their plan. Wander the area and observe their activity
- **Solution** -- Seven students should report to you and read the following numbers in any sequence
 - o Make a checkmark after each number is read
 - o If the number isn’t on the sheet, make a checkmark motion anyhow

- o After the last student has finished, if they don't have the correct answer, just tell them 'The shuttle doesn't respond' and let them go back for another try. You might want to amble along with one of the incorrect cadets to see what's going on in that subgroup of cadets

Sequence:

- 211 - 311 - 411 - 611 - 711 - 811 - 911
- These represent the docking expert, microbiologist, pilot, nav, life support expert, science officer, and mission specialist
- When time is up, stop them (or let them continue until they come up with a solution), collect the clues, and start the critique
 - o Let the cadets carry it as much as possible

MOONBASE ALPHA

Scenario Layout

Bubble 4: capacity 3	Bubble 5: capacity 5
Bubble 3: capacity 3	Bubble 6: capacity 3
Bubble 2: capacity 3	Bubble 7: capacity 3
Bubble 1: capacity 5	Bubble 8: capacity 5

MOONBASE ALPHA

Cadet Handout

Purpose: This exercise requires you to determine and consolidate critical information and act on it, under limitations of time and communication. All participants must know the plan and understand the group goal if you are to succeed. This will be a test of your individual and group problem-solving and communication skills.

Objective: You are the survivors of MOONBASE ALPHA who must get a critical bioprocessing unit delivered to L-5 colony Grissom Station. You must hurry because the part is necessary to prevent total collapse of the life support system, which was sabotaged. Sabotage also damaged your base and you are the only ones left. There is only one shuttlecraft left operational in the hangar bay. You must decide on the crew, decipher the access code, and deliver that part. One of you knows the secret to the shuttle code. None of you know each other well; you worked in different parts of the base. The shuttle can only hold a maximum crew of seven. The bio unit needs special care. You all have different pieces of information. Your time is running out. The L-5 colonists are dying.

The Task: Cadet _____ will act as the shuttle. When you're ready to blast off, your shuttle crew should report to "the shuttle" and state the access code. After you've stated the entire code, "the shuttle" will indicate whether or not you are on your way. Don't waste time with guesses; there are millions of possible combinations and the colonists are dying. The information slips you receive give your starting bubble location. Damage to the base has isolated you from each other and there is no surviving meeting area that can hold more than five at a time. Most areas can hold only three at a time. You have bio-suits enough for unlimited trips between areas, but you must quickly enter each bubble to avoid the ambient radiation outside the shelters. You may not give or show your info slips to anyone else; the knowledge is supposedly in your heads and must therefore be passed by word of mouth. Talk low enough that you cannot be heard in adjacent areas; after all, there's a vacuum between you.

As you start, only one of you even knows the form the access code will take.

Each of you will have an info slip of the following type:

- Example: You are Moreau
 - o Serial number 875
 - o You are the codes expert
 - o You've given the pilot all the codes needed for approach clearance
- Example: You are Falkenburg
 - o Serial number 837
 - o You are a plumber
 - o Bubble 5 has a slow leak and will be out of air in 2 days
- Example: You are VanRijn
 - o Serial number 419
 - o You are a neurologist

- o The code specialist is needed for approach verification at Grissom
- o

MOONBASE ALPHA - Info Slips

You are Solo Bubble 1
 Serial number 1011
 You are the cargo master
 Docking with Grissom will require an expert at docking

You are VanRijn
 Bubble 2
 Serial number 1111
 You are the cryogenics expert
 To stay healthy the bio unit will need constant care by the microbiologist

You are Scott Bubble 3
 Serial number 511
 You are the flight engineer
 The pilot must be on the crew

You are Kinneson Bubble 4
 Serial number 411
 You are the pilot
 The science officer will be needed to install the unit at Grissom Station

You are Dumarest Bubble 5
 Serial number 611
 You are the astronavigator
 You have a family at Grissom Station

You are Rogers Bubble 6
 Serial number 211
 You are the docking expert
 The mission specialist must help you during the complex docking procedures at Grissom Station

You are Spock Bubble 7
 Serial number 811
 You are the science officer
 You are also qualified as a copilot

You are McCoy Bubble 8
 Serial number 236
 You are the medical doctor
 The science officer also has an M.D.

You are Moreau Bubble 2

Serial number 875
 You are the codes expert
 You've given the pilot all the codes needed for approach clearance

You are Falkenburg Bubble 3
 Serial number 627
 You are the plumber
 MOONBASE ALPHA only has water for 2 weeks

You are Cabot Bubble 4
 Serial number 911
 You are the mission specialist
 You've done simulator training of just such mission

You are Kanobi Bubble 5
 Serial number 111
 You are the fuels specialist
 The access code consists of the serial numbers of the crew -- in any sequence

You are Carter Bubble 6
 Serial number 711
 You are the life support expert
 The crew must have a navigator for the in-flight corrections
 The auto systems are on the fritz

You are O'Leary Bubble 7
 Serial number 999
 You are the entertainer
 The USO return ship will be here in one week

You are Hollander Bubble 8
 Serial number 36
 You are the physical therapist
 You also are a computer specialist

You are Flandry Bubble 1
 Serial number 435
 You are the security guard
 The saboteur was killed in the blast

You are Talbot Bubble 2
 Serial number 23
 You are the archaeologist
 There are no monoliths on the moon

You are di Griz Bubble 3

Serial number 123
 You are the astrophysicist
 The next meteor shower will occur in 3 weeks

You are Forbin
 Serial number 566
 You are the computer specialist
 You already programmed the shuttle for the flight

Bubble 4

You are Atreides
 Serial number 47
 You are the visiting VIP
 You do not want to stay behind on the moon

Bubble 5

You are Prescott
 Serial number 666
 You are the chaplain
 You also have been trained in fuels and plumbing

Bubble 6

You are Cook
 Serial number 57
 You are the cook
 There it nothing to cook, only high vitamin pastes remain

Bubble 7

You are Samms
 Serial number 678
 You are the radiologist
 No refueling will be needed en route

Bubble 8

You are Xiang
 Serial number 900
 You are the Chinese exchange officer
 You want to find the language expert to find out what is going on here

Bubble 1

You are Gleen
 Serial number 849
 You are the psychologist
 The pilot was suffering from shock but appears to have recovered

Bubble 1

You are Carpenter
 Serial number 573
 You are the language expert
 All MOONBASE ALPHA personnel understand English even though they come from eight different countries

Bubble 5

You are Gordon
 Serial number 25
 You are the dietitian
 Bubble 5 has cracks and is slowly leaking

Bubble 8

You are Retief
Serial number 943
You are the intel officer
You believe the Iraqis are behind the bombing

Bubble 5

You are Piaget
Serial number 58
You are the education specialist
People learn 37 percent faster in lunar gravity

Bubble 8

CRITIQUE/OBSERVATION GUIDE

Planning:

- Who was the first to ask the question “What’s the task/problem here?”
- Who did organizing functions in the group?
- Who served as mediator in arguments?
- Who sketched the problem out so all could see or who centrally posted the ideas?
- Who pointed out some of the info is unneeded?
- Who served as recorder for the group?
- Who spoke first? Was what said pertinent?
- Who started or joined subgroups on the side during planning? What effect did it have on group progress?
- Who had the look of “I’ve got it” and either didn’t say anything, wasn’t listened to, or tried inappropriately to enter the conversation?
- Who served as gatekeeper, getting other people listened to and involved?
- Who kept the group on track?
- Who was time conscious?
- Who forced the group to make decisions and come up with a plan?
- Who realized/proposed a travel plan for getting the information around to the different “bubbles”?
- Who proposed they select the best brains to gather the data and meet at the larger “Bubble” to consolidate it?
- Who proposed they have backup plans?

Execution:

- Who served as field general, keeping the plan going?
- Who did what during crises (such as needing a new plan on the spot)?
- Who served as helper to explain to those who are not in the know?

- Who blew up at self, group, you, or other individuals?
- Who served as primary problem solver(s) on the field?
- Who goofed and how did others react?
- After action Q & A:
 - Whose participation was most helpful?
 - What did they do that was helpful?
 - Who helped most in the planning phase?
 - Who helped most in the execution phase?
 - What actions by whom helped solve the problem?
 - Who helped you most to understand what was going on? How?
- If you could do the task over:
 - What would you personally do differently?
 - What would you have others do differently?